

Research Methods — Guest Lecture 2 Notes

Literature review purpose · Reading strategy · Interpretation · Grey literature · Finalising research questions

Part 1

The real purpose of a literature review

A literature review is not a history lesson and it is not a summary of articles. It is the foundation that justifies every decision you make in your research — your question, your methodology, your data choice, and your interpretation.

What a literature review actually does

- Gives you a foundation of knowledge in your field — you cannot be a credible researcher without knowing what the leaders in your field are already doing
- Helps you position your research — it shows where your work fits relative to what already exists
- Justifies your research question — it proves that a gap or limitation exists that your study is addressing
- Provides methodological grounding — by reading other studies you discover which methods work for the type of problem you are tackling
- Supports your story — the literature is the evidence you use to tell the narrative of your proposal
- Validates your interpretations — when you draw a conclusion, the literature provides the backing

What a literature review is NOT

- Not a chronological history of the topic — you are not writing a timeline of who said what and when
- Not a list of summaries — do not describe each paper one by one. Group, synthesise, and argue
- Not a standalone section — it feeds directly into your problem statement, aim, and methodology
- Not something AI can do for you — AI gives you the most popular papers, not the most rigorous or most relevant ones for your specific question
- Not a place for casual language, first person ('I think...'), or opinion — academic voice only

The literature review exists to help YOU tell YOUR story. You use other people's work as evidence to support the argument you are building — not to replace your own voice.

Part 2

Reading strategy and organising your literature

You cannot read every paper in full — and you should not try. Use a structured reading strategy to quickly identify which papers are worth your time, then organise everything systematically from day one.

Three-step rapid evaluation — from the lecture

- Step 1 — Abstract: what are they claiming? What is the main argument? Does this relate to your topic?
- Step 2 — Conclusion: what did they actually find? Does it support or challenge your direction?
- Step 3 — Limitations: what could they NOT do or answer? This is your most valuable section — these limitations become your research questions
- If all three pass, THEN read the full paper closely — methodology, results, references
- If any step fails relevance, move on — do not waste time on papers that do not serve your study

How to organise your reading — practical tools

- Use a spreadsheet (Excel or Google Sheets) — one row per paper, columns for: reference, problem statement, methodology, results, conclusion, limitations, your remarks
- This is identical to the Literature Review Matrix on Moodle — fill it in as you go, not at the end
- Use referencing tools like Mendeley or Zotero — they automatically format references in Harvard style and keep your library organised
- Google Scholar: make it your starting point. Check citation counts — a highly cited paper has been validated by the academic community
- Higher citation count = more people have read, used, and trusted that paper. It is not a perfect measure but it is a useful signal
- DBS Library: always check here first for access. Never pay for a paper — use Sci-Hub with the DOI if needed

THE SPREADSHEET APPROACH — FROM THE LECTURE

The lecturer started their PhD by printing papers and creating a spreadsheet to track every source. The columns covered: reference, key arguments, methodology, limitations, and personal notes. This approach does two things: it stops you losing track of what you have read, and it forces you to critically engage with each paper rather than just passively reading it. The lecturer offered to share their spreadsheet template — ask for it if you have not received it.

Literature is always being published. You cannot claim to have read everything — and at Master's level, you are not expected to. What you CAN do is be thorough within a defined scope and acknowledge that newer work may exist. This is normal academic practice.

Part 3

The funnel approach — broad to specific

You will not find a paper that asks exactly the same question you are asking, in exactly the same way. Start broad, read widely, and let the reading narrow your focus naturally.

How the funnel works in practice

- Start with a broad area — e.g. renewable energy, fraud detection, healthcare AI
- Read widely across that area — look for recurring themes, debates, and limitations
- As you read, your topic narrows — you identify a specific angle, domain, population, or method that has not been fully explored
- That narrow focus becomes your research problem and eventually your research question
- Topics CAN and DO change as you read — this is normal. Do not commit too early
- Students who say 'I cannot find papers on my topic' are usually searching too specifically. Start broad, then narrow

EXAMPLE OF THE FUNNEL IN ACTION — FROM THE LECTURE

Broad start: 'Renewable energy' After reading: 'Offshore wind energy in Ireland' After more reading: 'Risk management in offshore wind energy development in Ireland' Research problem identified from limitations: 'Existing risk models do not account for Irish coastal weather variability' Research question: 'How effective are existing risk management frameworks for predicting operational downtime in Irish offshore wind installations between 2018 and 2023?' The question was not decided before the reading — the reading produced the question.

Part 4

Interpretation — what makes your research yours

Two researchers with identical data and identical information will interpret it differently. That interpretation — shaped by your choices, your background, and your reading — is what makes your research original.

THE X-RAY ANALOGY — FROM THE LECTURE

Imagine two doctors both looking at the same X-ray of a broken leg. The only thing they can agree on with certainty: it is an X-ray. It shows a bone. There is a fracture. After that — one doctor says: operate immediately. The other says: put in a cast, give antibiotics, give it four weeks. Same information. Different training, different experience, different interpretive frameworks. Now apply this to research: 10 students in the same class given the same dataset will ask 10 different questions, apply 10 different methods, and reach 10 different conclusions. That is not a problem — that IS research. Your interpretation of the data and the literature is what makes your work uniquely yours.

THE BUS STOP ANALOGY — FROM THE LECTURE

Everyone at the bus stop gets the same piece of information: 'The next bus is 45 minutes away.' One person goes for coffee. One walks to the next stop. One calls a taxi. One decides to wait. Same information — completely different decisions based on individual circumstances, priorities, and interpretations. In research, the 'information' is the data and the literature. Your interpretation of it — guided by your reading — is your unique contribution.

What this means for your dissertation

- Your literature review gives you the framework to interpret data — that framework is what you have built through reading
- You do not need a unique dataset or a groundbreaking discovery — you need a rigorous, evidence-based interpretation
- Even if someone else has studied the same topic, your data, your timeframe, your location, your methodology, or your angle will differ — making your contribution valid
- The ability to explain your work clearly and justify every decision is what makes research replicable and credible
- Research is replicable when someone else can follow your methodology and get a comparable result — that is the standard you are aiming for

Part 5

Gap vs limitation — a critical distinction

Many students say they are 'closing a gap' in the literature. This is the wrong framing. You are addressing a limitation — and the distinction matters both academically and practically.

Term	What it implies	Why it is problematic or correct
Gap	Something is entirely missing — a complete absence in the literature	Almost never true at Master's level. Claiming to close a gap implies you have read everything and nothing exists. New papers are published constantly. You cannot close what keeps opening.
Limitation	A specific constraint in an existing study — something the authors acknowledge they could not do	Always true and always defensible. You are building on published work, addressing what the authors themselves said was incomplete. This is the correct framing.

How to frame this correctly in your proposal

- Wrong: 'There is a gap in the literature regarding X, which this study will fill'
- Correct: 'According to Smith and Jones (2022), a key limitation of their study was the absence of X. This study seeks to address that limitation by...'
- Wrong: 'No research has been done on this topic'
- Correct: 'Existing research on this topic has focused primarily on [region/population/method]. This study extends that work by applying [new approach] to [new context]'
- Every two seconds a new paper is published somewhere in the world. You cannot claim completeness — only thoroughness within your defined scope

Part 6

Grey literature — what it is and how to use it

Grey literature refers to reports, white papers, policy documents, and data publications produced by organisations outside the traditional academic peer-review process. It is acceptable and expected — but must not dominate your literature review.

Grey literature source	Examples	How to use it
International organisations	WHO, World Bank, IMF, United Nations, OECD, UNESCO	Use for statistics, global reports, and policy context. Cite as you would any source.
Government bodies	Central Statistics Office (Ireland), European Commission, UK ONS, US Census Bureau, data.gov.ie	Use for national/regional data, legislation, and official statistics. Highly credible for applied research.
Industry reports	Gartner, McKinsey, Deloitte, PwC, Accenture, IBM research	Use sparingly — they are not peer-reviewed but provide real-world context. Do not let them replace academic sources.
Regulatory and standards bodies	GDPR (EU), NIST (cybersecurity), IEEE standards, ISO standards	Use when discussing compliance, data governance, or technical standards relevant to your study.
Previous dissertations (DBS thesis repository)	MSc Data Analytics and MSc Business Analytics theses	Use as structural models and to find datasets and limitations to build on. Cite them properly.

Grey literature should support your academic sources — not replace them. If your literature review is dominated by industry reports and government documents with few peer-reviewed journal articles, it will not pass academic scrutiny.

Part 7

Direct quotes vs paraphrasing — and how to cite

In academic writing, you almost always paraphrase rather than quote directly. But whether you quote or paraphrase, you must always give credit to the original author — this is what Harvard referencing is for.

Approach	When to use	Example
Paraphrase (preferred)	When you want to use an author's idea or finding but express it in your own words. This is the standard approach in academic writing.	According to Rajiv (2022), data centres in Ireland account for a significant proportion of national energy consumption — a figure that has increased substantially in recent years.

Direct quote (rare)	Only when the exact wording matters — e.g. a legal definition, a very precise statistical claim, or a phrase that cannot be rephrased without losing meaning.	Rajiv (2022, p.14) states that 'data centres currently consume 25% of Irish national energy output'.
Synthesis (strongest)	When you combine findings from multiple sources to build your own argument. This demonstrates the highest level of critical engagement.	Multiple studies confirm that energy consumption from data infrastructure is a growing concern in Ireland (Rajiv, 2022; Smith, 2021; O'Brien, 2023), suggesting that sustainability must be a central consideration in cloud infrastructure planning.

Harvard referencing — key rules

- In-text citation: According to [Author surname] ([year])... OR at the end of a sentence: ...(Author surname, year)
- Never use 'they', 'he', 'she' without naming the source — always use the author's name
- Never use 'I found', 'I think', 'In my opinion' — this is an objective academic document
- Reference list at the end: full details in alphabetical order by author surname — title, journal, volume, issue, pages, DOI
- Be consistent: do not mix Harvard in-text citations with IEEE numbered references. Pick one and use it throughout
- Plagiarism in reference lists is not penalised — copying a citation exactly is expected and correct

Part 8

Finalising your research question — complex and arguable

The final two criteria for a strong research question: it must be complex enough to require investigation, and it must be arguable — meaning it opens space for debate and future research rather than closing it.

Criterion	What it means	Bad example	Good example
Complex	Cannot be answered with a simple yes or no. Requires original data, analysis, and argument to answer.	Has homelessness increased in the UK?	What are the economic and political factors that have contributed to patterns of homelessness in the UK over the past decade?
Arguable	Leaves room for debate, further research, and extension by others. Research is a continuous process — your question should open a door, not close one.	Is machine learning used in fraud detection? (Yes — done.)	How do ensemble learning models compare to deep learning approaches in detecting real-time credit card fraud in European banking datasets?

How to test if your question is complex and arguable

- Can it be answered with yes or no? If yes — reframe it. Add specific factors, populations, timeframes, or measurable variables
- Can it be answered with a quick Google search? If yes — it is not a research question. Research questions require original analysis to answer
- Does it leave open room for someone else to continue the work? Good research opens doors — it does not claim to have shut them
- Does it require you to engage with data, analyse it, and interpret the results? If yes — you are on the right track
- Does answering it take the form of an argument supported by evidence? That is the test for an arguable question

MAKING A QUESTION COMPLEX AND ARGUABLE — WORKED EXAMPLE

Weak: 'Is AI used in healthcare?' → Yes. One word answer. Not a research question. Better: 'How is AI used in healthcare diagnosis?' → Descriptive. Better, but still not requiring much analysis. Strong: 'How effective are convolutional neural network models in detecting early-stage lung cancer in CT scan data compared to traditional radiologist diagnosis in Irish hospital datasets (2019–2023)?' Why it works: Specific method: CNN models Specific domain: lung cancer detection Comparison point: vs radiologist diagnosis Specific dataset context: Irish hospitals Measurable variable: detection effectiveness Timeframe: 2019–2023 Cannot be answered yes or no Opens space for future research: other cancer types, other model architectures, other countries

Quantitative research approach — clarification from the lecture

- MSc Data Analytics students primarily use a QUANTITATIVE approach — working with measurable, numerical data
- Qualitative data (text, categories) CAN be converted to numbers for analysis — e.g. male/female becomes 0/1
- Mixed methods (quantitative + qualitative) is technically allowed but NOT recommended in 12 weeks — it doubles the workload
- The 5–10 year rule applies to ARTICLES only — your dataset can be older if the model or methodology still applies
- Your research question will continue to improve in Applied Research Methods when you finalise your dataset — do not wait for perfection now

Things every speaker agreed on

- Read. There is no shortcut. Even two articles on the bus is a win — use the reading strategies you have been taught
- Follow your passion — if you care about the topic you will do better work and speak about it more confidently
- Look at conclusions and limitations sections first — that is where your research questions come from
- There is no perfect answer. You are contributing to knowledge, not completing it
- Google Scholar is your best friend — check citation counts as a proxy for credibility
- Make your literature review matrix your daily working document — do not leave it to the end
- Use referencing tools (Mendeley, Zotero) from day one
- Your voice must be consistent throughout — AI output breaks the tone. Examiners will notice
- Address a limitation, not a gap — the distinction matters
- Small, focused, replicable contributions are the goal. Not world-changing discoveries

Reading week action list

- Complete your literature review — minimum 20 peer-reviewed articles, grouped by theme or methodology
- Fill in every row of your Literature Review Matrix — especially the limitations column
- Draft your problem statement — based on a limitation identified in the literature
- Draft your aim — what you want to achieve in response to that limitation
- Draft your primary research question — specific, focused, measurable, complex, and arguable
- Confirm data availability — in parallel with your question, verify that real public data exists for your topic
- Set up Mendeley or Zotero — get your references in order now so you are not doing it the night before submission